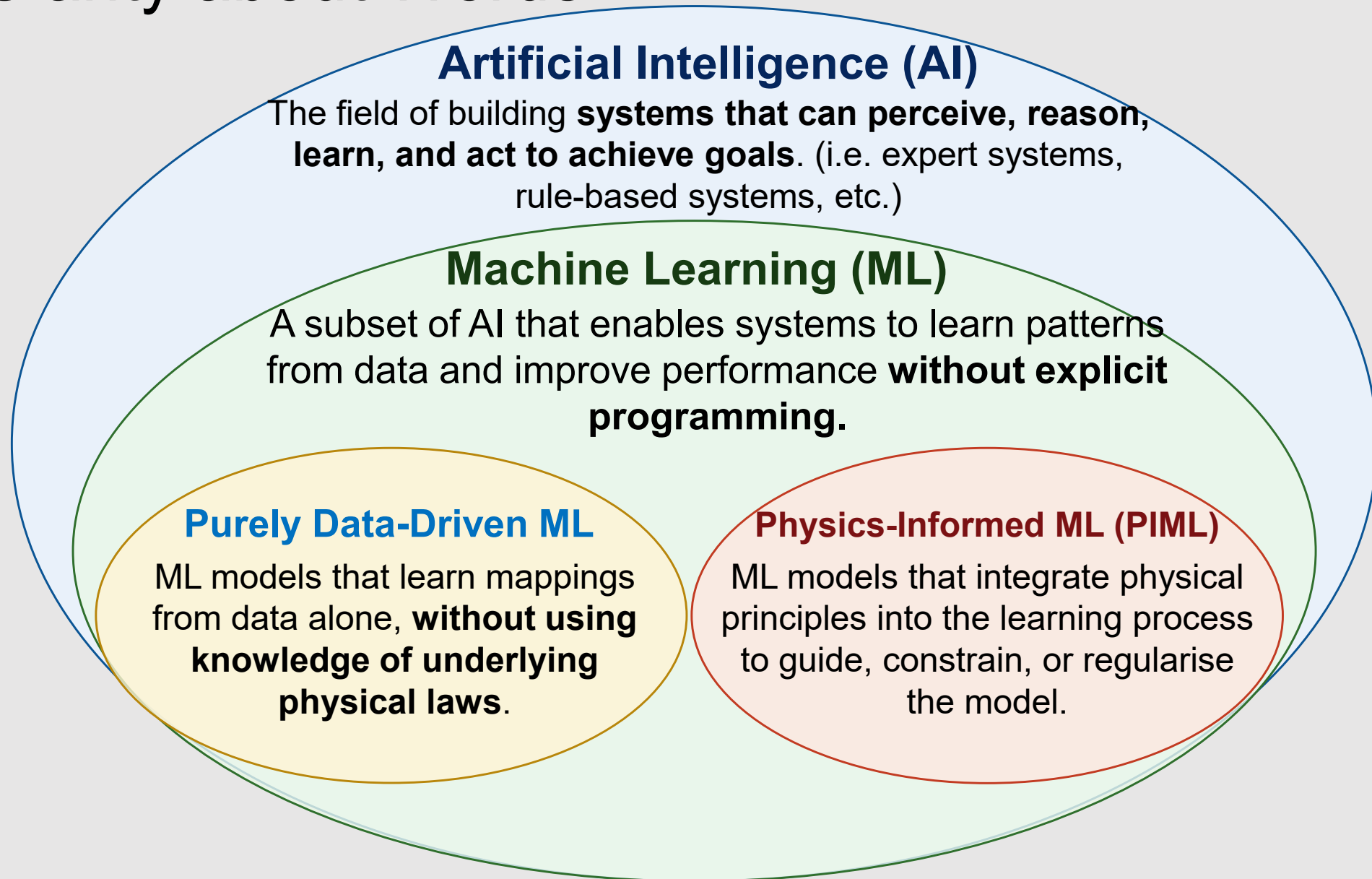


Overview of ML methods

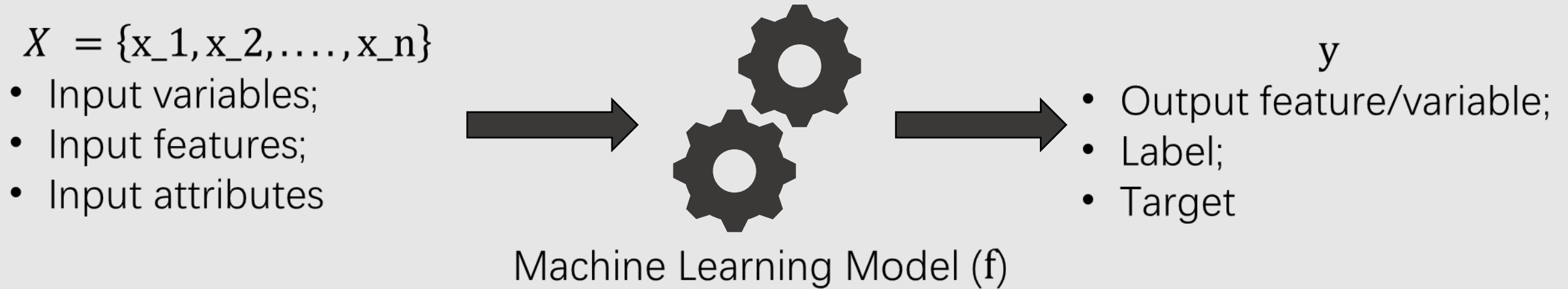
From NTH problems to supervised machine-learning tasks

Some Clarity about Words



What is a Machine Learning Algorithm?

A machine learning algorithm is a set of **rules and mathematical instructions** used by computers to find patterns in data.



$$y = f(\beta, X)$$

Model Parameters

Model Learning is to adjust the model parameters β throughout a number of iterations to maximize/minimize a likelihood/error function on y .

Machine Model Learning/Training

- The model learns β or is trained during the learning/training process using the sampled observations of y to reproduce the right y at unknown locations without extra programming;
- That is why machine learns;
- Many different approaches for three ways of learning:
 - Supervised learning;
 - Unsupervised learning;
 - Semi-supervised learning.

Three Ways of Learning/Training

Learning approaches	Supervised Learning	Unsupervised learning	Semi-supervised learning
Definition	• $X=(x_1,x_2)$ and $y = \{\text{label1}, \dots, \text{labeln}\}$ or $y \in \mathbb{R}$; • A training set with many examples of (X, y); • The model learns on the examples of the training set to produce the right value of y for an input vector X	• $X=(x_1,x_2)$ and $y = \{\text{label1}, \dots, \text{labeln}\}$; • A training set with many examples of (X, y); • The model learns to group the examples X of the training set based on similarity (closeness) or probability	• $X=(x_1,x_2)$ and $y = \{\text{label1}, \dots, \text{labeln}\}$; • A training set with many examples of (X, y), with some of (X, y); • The model labels the data in the training set using a modified unsupervised learning procedure
Typical tasks	Regression; Classification	Clustering, Dimensionality Reduction, Density Estimation	Classification; Anomaly or fraud detection
Examples	Tree-based method; GPR; NNs, etc.	k-Means, PCA, Autoencoders, Gaussian Mixture Models, etc.	Self-training / pseudo-labeling; Co-training, etc.

Four task Framings of Supervised Learning

Regression

Predict a continuous value
CHF, pressure drop, HTC, etc.

Classification

Predict a label or state
Safe/Unsafe, Flow Regime, Accident Class

Surrogate model

Emulate an expensive solver
CFD output approximation

Decision support

Use model outputs carefully
design search, margin screening, operator aid

Main focus today: supervised regression and surrogate modelling for tabular engineering data.

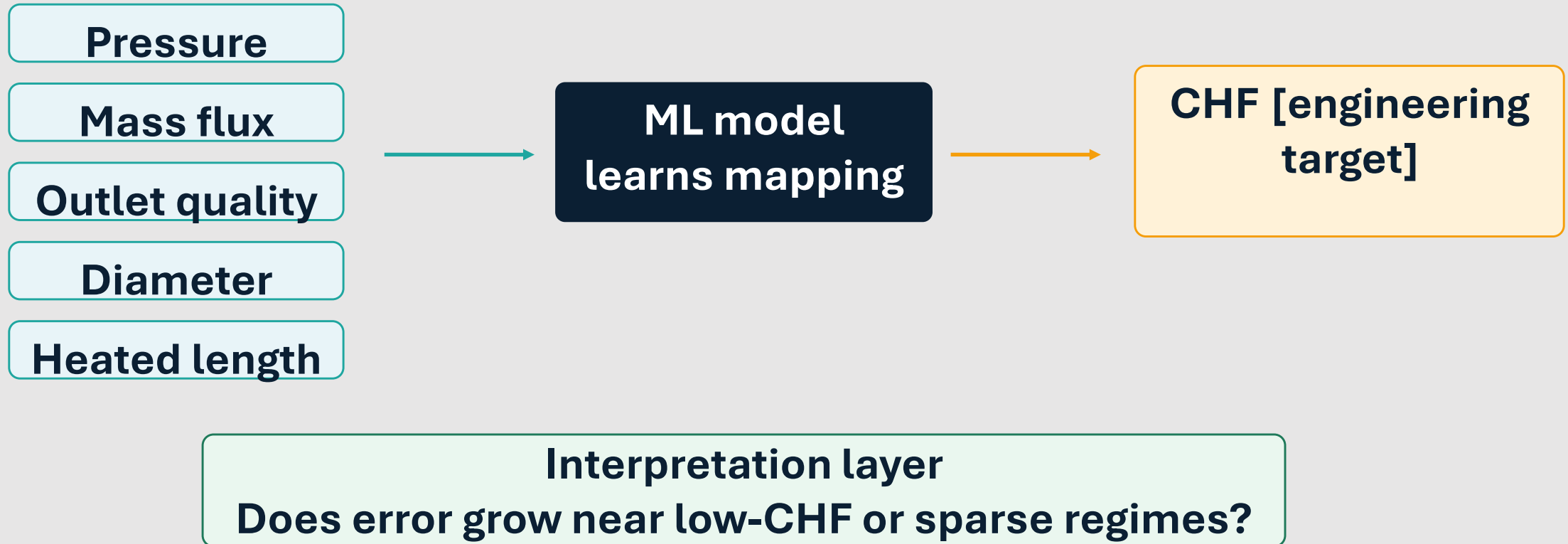
Mapping engineering questions to supervised learning

Engineering question	ML framing	Target
Predict CHF from operating conditions	Regression	CHF
Estimate pressure drop from simulation data	Regression / surrogate	Δp
Detect flow regime from experiment features	Classification	regime label
Replace expensive CFD over a parameter sweep	Surrogate regression	field or QoI
Flag an operating point outside training range	Domain check	in/out domain

Formulate the problem first; choose the algorithm second.

Running case study: CHF prediction

Critical heat flux is a natural supervised-regression example because it is safety-relevant, data-rich in places, and validity-range sensitive.



Dataset anatomy: samples, features, target, metadata

A supervised ML dataset is a table with meaning attached to every column.

Number	Reference ID	Tube Diameter (m)	Heated Length (m)	Pressure (kPa)	Mass Flux (kg/m ² /s)	Outlet Quality	Inlet Subcooling (kJ/kg)	Inlet Temperature (degree C)	CHF (kW/m ²)	Reference name
8461	31	0.0108	2	6860	504	0.87	222	240.1	1046	Zenkevich
8462	31	0.0108	2	6860	504	0.88	125	260.16	996	Zenkevich
8463	31	0.0108	2	6860	995	0.62	219	240.74	1570	Zenkevich
8464	31	0.0108	2	6860	995	0.64	130	259.15	1477	Zenkevich
8465	31	0.0108	2	6860	1504	0.46	217	241.16	1895	Zenkevich

Metadata is often for grouping, provenance, or validation — not necessarily a normal input feature.

A “row” is not just data. It represents an experiment, simulation, or operating condition.

Features: X

Target: y

Feature definition is an engineering decision

The feature list is not neutral: it encodes assumptions about physics, measurement, and deployment.

- **Availability** Would this variable be known at prediction time?
- **Units and scaling** Are MPa/bar, $\text{kg m}^{-2} \text{s}^{-1}$, mm/m consistent across sources?
- **Physical meaning** Should we use raw variables, dimensionless groups, or both?
- **Redundancy** Are quality, subcooling, and inlet temperature over-specifying the same state?
- **Provenance** Is source/facility metadata a feature or a validation group?

Bad feature choice can create leakage, spurious correlation, or invalid extrapolation

Target definition: what exactly are we predicting?

The target is not just a column name. It is the engineering quantity, context, time horizon, and measurement convention.

Scalar Qol:
CHF, pressure drop, HTC, PCT

Class label:
**safe/unsafe, flow regime,
accident type**

Field output:
**temperature, velocity, void
fraction field**

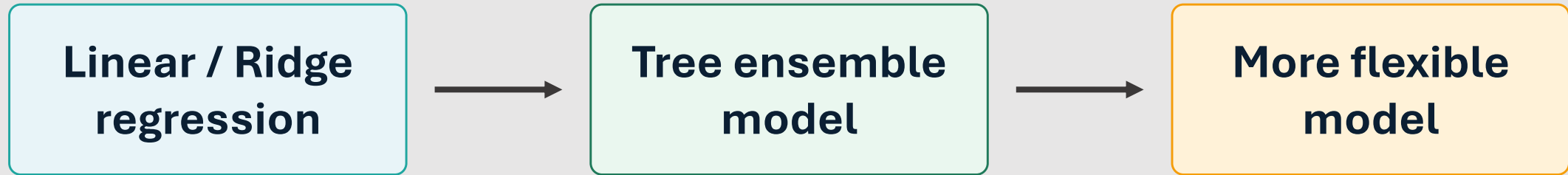
Time horizon:
**next 10 s, peak over transient,
final state**

**Common
mistake**

**Training against a target that is inconsistent across
sources, regimes, or measurement definitions.**

Start with transparent baselines

For engineering ML, the first useful model is often not the most complex model.



- Establish a sanity check before tuning complex models;
- Reveal whether the signal is easy, nonlinear, or noisy;
- Make later improvements credible rather than cosmetic;
- Expose errors caused by features, units, or split design.

Method map for the first week

Linear / Ridge	transparent baseline	small-to-medium tabular data
Random Forest / Gradient Boosting	strong tabular workhorses	nonlinear interactions
Gaussian Process	surrogate + uncertainty	small data, adaptive sampling
Neural Network	flexible nonlinear mapping	larger data, disciplined validation
UQ / VVUQ layer	calibration and domain evidence	safety-relevant use

The lecture gives theory; later sessions/labs build skill.

Linear and Ridge regression: the first baseline

A transparent model asks: can a weighted combination of features explain the target?

$$\hat{y} = b + w_1x_1 + w_2x_2 + \dots + w_px_p$$

Linear regression: optimises the coefficients by minimizing the loss function (sum of squared errors (SSE));

Ridge regression: adds regularisation $\lambda \sum_{i=1}^p w_i^2$ in the loss function to discourages very large coefficients and often improves stability.

Good for:

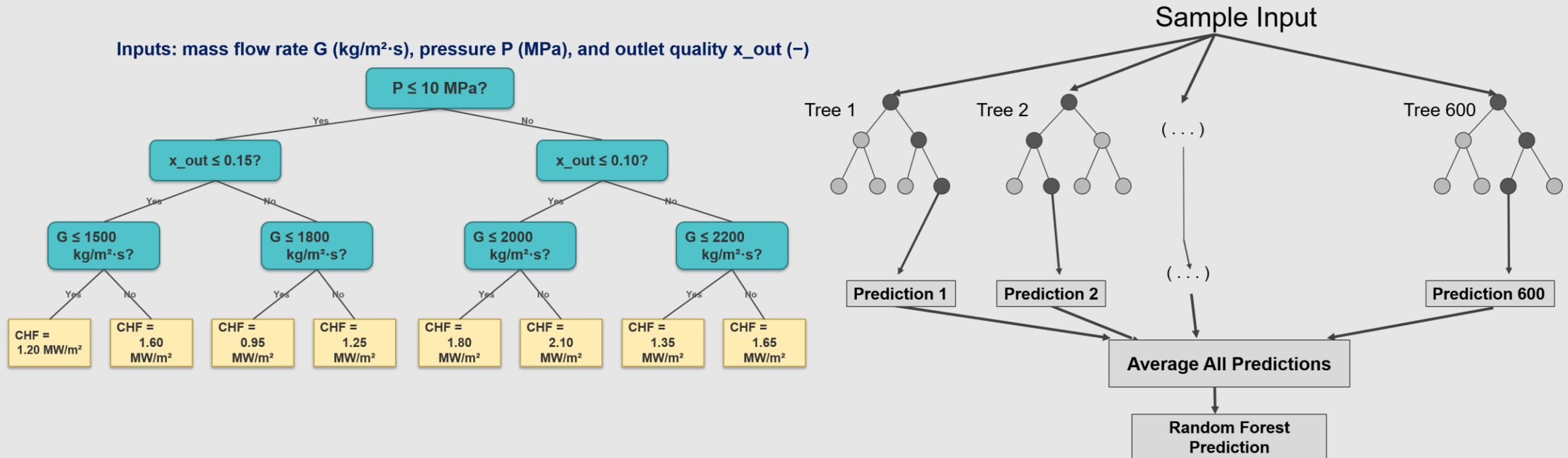
- sanity checks
- interpretable coefficients
- small data
- baseline lab workflow

Watch for:

- nonlinearity
- correlated features
- scaling sensitivity
- not automatically physical

Tree ensembles: strong tabular-data workhorses

Random forests and gradient boosting learn nonlinear rules by splitting the feature space.

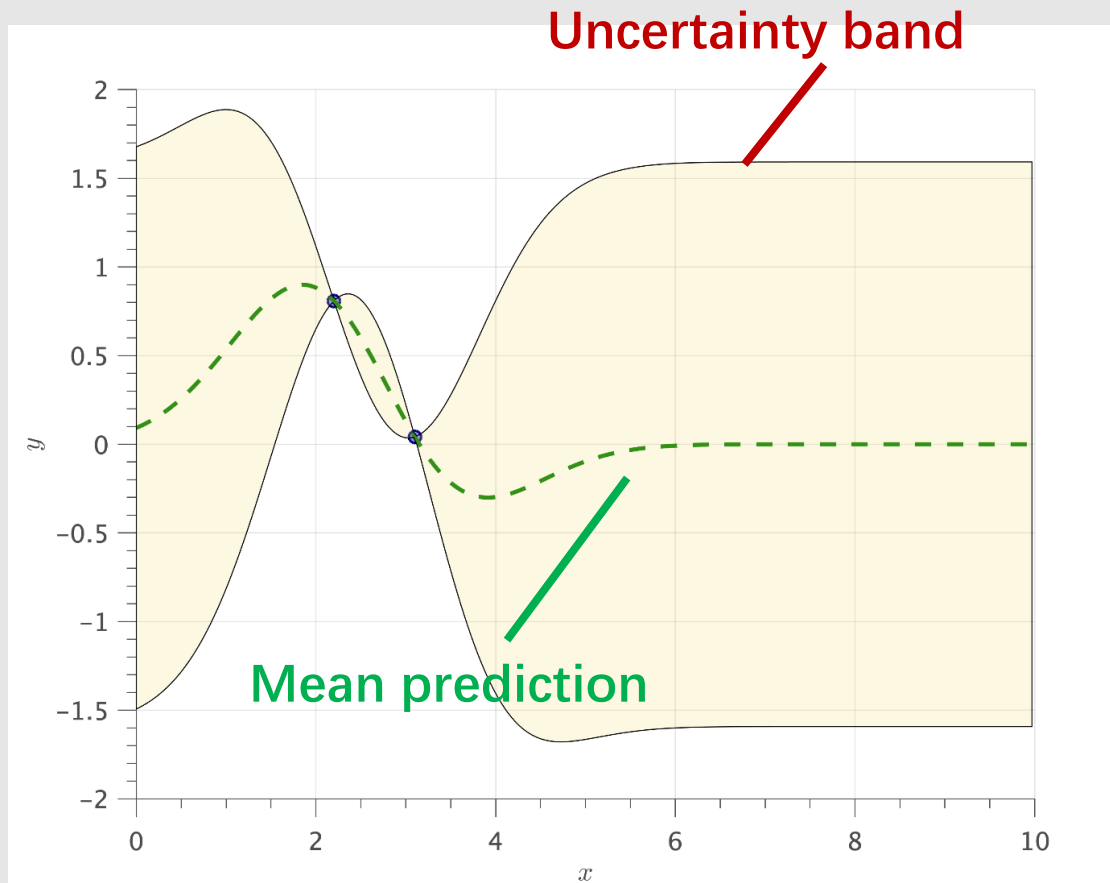


Strength: captures nonlinear interactions without heavy preprocessing.

Caution: still needs honest validation and physical diagnostics.

Gaussian Processes: surrogate modelling with uncertainty

A GP predicts a mean and an uncertainty band. That makes it attractive for expensive simulations and small-to-moderate datasets.



Useful when:

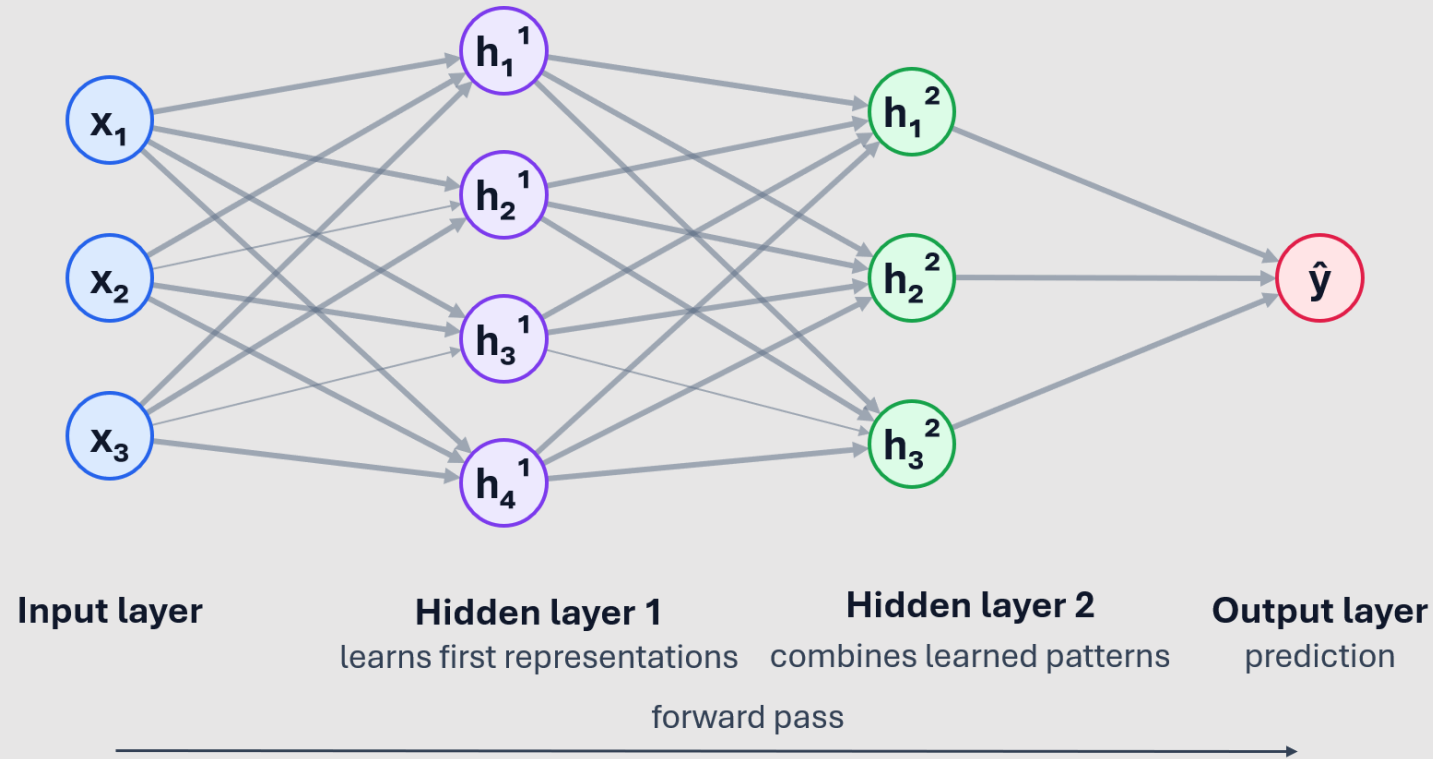
- each simulation is costly
- data are limited
- uncertainty guides new runs
- BEPU/UQ needs fast QoI evaluation

Limitations:

- scales poorly with very large data
- sensitive to kernel assumptions
- uncertainty is not magic OOD proof

Neural networks: flexible nonlinear models, not magic

Feed-forward neural networks are powerful function approximators for nonlinear mappings, including CHF and code-output surrogates.



Good for nonlinear response surfaces — but needs enough data, scaling, validation, and uncertainty awareness.

Note: More parameters $\neq$ more credibility

Method choice: match the model to the engineering use

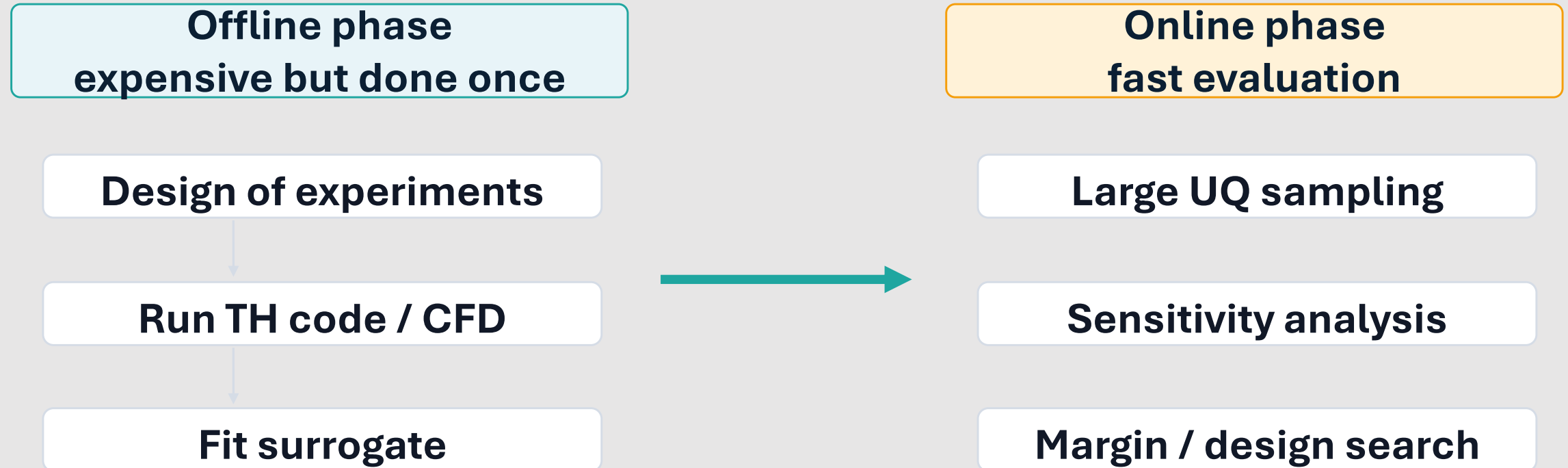
The “best” algorithm depends on data size, nonlinearity, interpretability, uncertainty needs, and deployment context.

Situation	Reasonable starting point	Risk to watch
small dataset	Ridge / SVR / GP	overfitting
smooth response	linear, PCE, GP	regime discontinuity
strong nonlinearity	trees / boosting / NN	source bias
need uncertainty	GP, ensemble, quantile model	miscalibration
need interpretability	linear, trees, SHAP + physics check	false explanation

Rule of thumb: use the simplest model that gives credible performance under the validation design you actually need.

Surrogate modelling in an NTH workflow

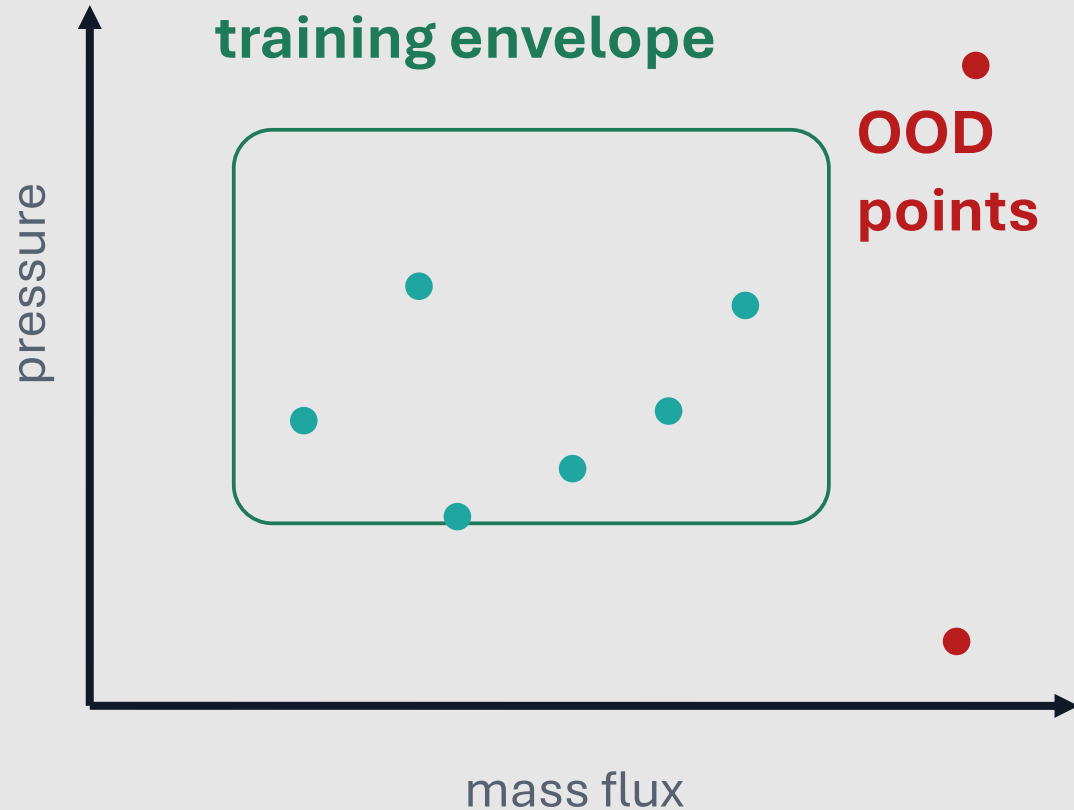
A surrogate is a fast approximation of a slow model. Its credibility depends on the domain where it was trained and tested.



**Validity envelope:
Where can this surrogate be trusted?**

In-domain vs out-of-domain use

ML models usually interpolate more safely than they extrapolate. NTH variables and regimes define the domain of use.



In-domain:

- inside sampled ranges
- similar sources/regimes
- error estimate meaningful

Out-of-domain:

- new regime/geometry
- different facility/source
- unsafe extrapolation risk

Lecture 1.3 will turn this into validation and failure-gallery examples.

What makes engineering ML different?

Average accuracy is only one part of credibility in safety-relevant thermal hydraulics.

- 1 Accuracy**
How large are typical errors?
- 2 Regime behaviour**
Where are the errors largest?
- 3 Physical plausibility**
Does the trend make engineering sense?
- 4 Uncertainty**
Are intervals calibrated and useful?
- 5 Domain of use**
What inputs and sources are valid?
- 6 Traceability**
Can another analyst reproduce it?

The question is not only “does it fit?” but “can we trust it for this engineering use?”

Before fitting a model: a five-question checklist

- 1 What is the engineering decision or question?
- 2 What exactly is the target?
- 3 Which features are available before prediction?
- 4 What data source or regime must the model generalise to?
- 5 What metric and diagnostic plots will reveal useful failure modes?

If these are unclear, algorithm choice is premature.

What we deliberately leave for later

This summer school is an orientation layer. You can go deeper once the supervised-learning frame is clear.

**Deep-learning architectures
CNNs, LSTMs, transformers**

**Unsupervised-learning taxonomy
clustering, autoencoders**

**Full Bayesian derivations
MCMC, posterior maths**

**PINN equations
loss balancing, collocation**

**Regulatory assurance cases
VVUQ evidence packages**

**Large-language models
not part of this core workflow**

**Today: enough ML vocabulary to start doing disciplined NTH
regression.**

Take-home messages

- 1 For this school, ML mainly means supervised mappings from features to engineering Qols.
- 2 Problem formulation comes before algorithm choice.
- 3 CHF prediction is a useful running example: pressure, mass flux, quality, geometry $\rightarrow$ CHF.
- 4 Simple baselines are not primitive; they are workflow sanity checks.
- 5 Engineering ML must state its domain of use, not only its average error.